RAJALAKSHMI ENGINEERING COLLEGE

Mahesh D 251001139-2025-IT

M2

| | |
|---|---|
| **Started on** | Friday, 31 July 2026, 12:15 PM |
| **State** | Finished |
| **Completed on** | Friday, 31 July 2026, 1:16 PM |
| **Time taken** | 1 hour |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

void function(int n)
{
    int c = 2;

    int i = 1;
    int s = 1;

    while (s <= n)
    {
        c++;

        i++;
        c++;

        s += i;
        c++;
    }

    c++;

    printf("%d", c);
}

int main()
{
    int n;
    scanf("%d", &n);
    function(n);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9 | 12 | 12 | ✔ |
| ✔ | 4 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahesh D 251001139-2025-IT

M2

| | |
|---|---|
| **Started on** | Friday, 31 July 2026, 12:32 PM |
| **State** | Finished |
| **Completed on** | Friday, 31 July 2026, 1:17 PM |
| **Time taken** | 44 mins 18 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
void func(int n)
{
    int c=0;
    if(n==1)
    {
      //printf("*");
      c++;
    }
    else
    {
        c++;
     for(int i=1; i<=n; i++)
     {
         c++;
       for(int j=1; j<=n; j++)
       {
           c++;
          //printf("*");
          c++;
          //printf("*");
          c++;
          break;
          c++;
       }
       c++;
     }
     c++;
   }
   printf("%d",c);
 }

 int main(){
     int n;
     scanf("%d",&n);
```

```
36      func(n);
37      return 0;
38  }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahesh D 251001139-2025-IT

M2

| | |
|---|---|
| **Started on** | Friday, 31 July 2026, 12:37 PM |
| **State** | Finished |
| **Completed on** | Friday, 31 July 2026, 1:17 PM |
| **Time taken** | 39 mins 55 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>

void factor(int n)
{
    int c=0;

   for (int i = 1; i <= n;++i)

   {
       c++;
    if (n % i== 0)
       {

         //printf("%d ", i);
         c++;
       }
       c++;
    }
    c++;

    printf("%d",c);
 }

int main(){
    int n;
    scanf("%d",&n);
    factor(n);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahesh D 251001139-2025-IT

M2

| | |
|---|---|
| **Started on** | Friday, 31 July 2026, 12:50 PM |
| **State** | Finished |
| **Completed on** | Friday, 31 July 2026, 1:17 PM |
| **Time taken** | 26 mins 37 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
void function(int n)
{
    int c= 0;
    int c1=0;
    c1++;

    for(int i=n/2; i<n; i++){
        c1++;
        for(int j=1; j<n; j = 2 * j){
        c1++;
            for(int k=1; k<n; k = k * 2){
                c1++;
                c++;
                c1++;
            }
            c1++;
        }
        c1++;
    }
    c1++;

    printf("%d",c1);
}

int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | 30 | 30 | ✔ |
| ✔ | 10 | 212 | 212 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahesh D 251001139-2025-IT

M2

| | |
|---|---|
| **Started on** | Friday, 31 July 2026, 1:10 PM |
| **State** | Finished |
| **Completed on** | Friday, 31 July 2026, 1:17 PM |
| **Time taken** | 7 mins 32 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
void reverse(int n)

{
    int c=0;
   int rev = 0;
   c++;
   int remainder;
   c++;
   while (n != 0)

    {
        c++;
        remainder = n % 10;
        c++;
        rev = rev * 10 + remainder;
        c++;
        n/= 10;
        c++;

    }
    c++;
//print(rev);

printf("%d",c);
}

int main(){
    int n;
    scanf("%d",&n);
    reverse(n);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahesh D 251001139-2025-IT

M2

| | |
|---|---|
| **Started on** | Sunday, 30 August 2026, 6:37 PM |
| **State** | Finished |
| **Completed on** | Sunday, 30 August 2026, 6:39 PM |
| **Time taken** | 1 min 34 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |